

Decision-Making Under Uncertainty: Prophet and Philosopher Inequalities

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Abstract

This note discusses known results regarding the polynomial-time approximability of the optimal online Bayesian bipartite matching algorithm and presents a new philosopher inequality for online Bayesian ski rental. First, known results of online Bayesian bipartite matching [PPSW21; BDP+24], online Bayesian ski rental [HJS24] and other variations are described. [PPSW21] considered for online Bayesian bipartite matching both the edge-weighted case and the edge-unweighted case and proved that approximating the optimal online algorithm is **PSPACE**-hard within some constant $\alpha < 1$. Also, [PPSW21] described a polynomial-time algorithm which approximates the optimal online algorithm within a factor of 0.51, beating the best possible prophet inequality. The latter result was subsequently improved, with the current known approximation ratio being 0.678 [BDP+24]. Finally, we show a new linear program that captures the optimal online Bayesian ski rental algorithm *precisely*, similar to the known result regarding the polynomial-time computability of the optimal online Bayesian bipartite matching with a single bin [PPSW21].

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1 Introduction

Decision theory initially focused on instantaneous decisions, but later gave us stopping rules and the gem of prophet inequalities [KS78]. Later, the internet age brought us new business models relying exclusively on stochastic decision making- online advertising, ride hailing, kidney exchanges- in which the changing environment affected by the agents' decisions can be often abstracted as an evolving weighted bipartite graph. On the other hand, there are various real-world applications in which the agents' decisions capture the classic rent/buy dilemma, e.g., whether to rent/buy an expensive equipment or a luxury item with unknown number of days of utility, networking/scheduling problems where there are multiple servers with different service guarantees and prices. Here we cover two problems of these types in which the setting is online and Bayesian, the online Bayesian bipartite matching problem, and the online Bayesian ski rental problem.

1.1 Problem Statements

Online Bayesian Bipartite Matching. The input to this problem is a random bipartite graph, revealed over time. Initially, the m nodes on one side of the graph, termed bins, are present. The n nodes on the other side, termed balls, are revealed over time, in a fixed order known to us. Initially, we know for each ball t the probability p_t of it actually arriving, as well as the non-negative weight $w_{i,t}$ of the edge connecting it to any bin i - if it arrives. If ball t does not arrive, we do nothing at time t ; if it does arrive, we can choose to match it, irrevocably, to some unmatched neighbor i before time $t + 1$, yielding a profit $w_{i,t}$. Our goal is to maximize the overall expected profit.

Online Bayesian Ski Rental. The input to this problem is a random sequence of requests, revealed over time. The T requests, termed days, are revealed over time, in a fixed order known to us. Each of the T days has good or bad weather. Initially, we know for each day t the probability of it arriving as a good weather day. Every day after observing the weather, we have to make a decision of whether to continue or stop (i.e, rent or buy), irrevocably. If we stop, we suffer a final cost of M , whereas if we continue, we suffer a cost of 1 if the weather was good, while we suffer a cost of 0 if the weather was bad. Our goal is to minimize the overall expected cost.

1.2 Related Work

The literature on online Bayesian selection problems is quite extensive. We briefly mention some of the most relevant work here. See also a survey on online Bayesian bipartite matching [HTW24]. Many online Bayesian selection problems have been intensely studied over the years via the lens of competitive analysis, or *prophet inequalities*, so-called as they compare with a "prophet" who has foresight of the realized input, and computes the optimal solution up front. A fundamental result in the stopping theory literature, the first prophet inequality, a $1/2$ -competitive algorithm for the single-item online Bayesian selection problem, was first given in the late 70s [KS78]. Multiple algorithms achieving this bound are known [Sam84; KW12]. A competitive ratio of 0.5 is known to be achievable via myriad approaches [FGL14; EFGT20; DFKL20], including pricing-based ones [FGL14; DFKL20], and this is tight even without restricting to pricing-based algorithms, and even with a single bin, or single item to sale [KS78]. For the unweighted and vertex-weighted problem, online Bayesian bipartite matching algorithms yield an approximation ratio of 0.666, but no more than 0.75 [TWW22]. The special case of i.i.d. distributions at all time steps has also been extensively studied, among others also in [TWW22]. Papadimitriou et al. [PPSW21] pioneered the study of the online Bayesian bipartite matching problem via the lens of polytime (online) approximation of the optimal *online algorithm*. The optimal online algorithm lacks foresight about the future, but

possesses unlimited computational power, or time to "think". This optimal policy is fittingly called a "philosopher", and approximation of this policy's value by polytime online algorithms are referred as *philosopher inequalities*. [PPSW21] showed that unless $\mathbf{P} = \mathbf{PSPACE}$, no α -approximate philosopher inequality is possible for some constant $\alpha < 1$. In contrast, [PPSW21] showed that the best philosopher inequality yields a higher approximation ratio than the best-possible prophet inequality's competitive ratio of 0.5. The 0.51 bound given by [PPSW21] was sequentially improved [SW21; BDM22; SW+23; BDP+24], with the current best known approximation being 0.678 [BDP+24]. In addition, [BDP+24] showed a 0.685-approximation of the online optimum for a vertex-weighted special case, exceeding the 0.666-approximation of the offline optimum obtained by Tang et al. [TWW22].

The ski rental problem has been looked at from a distributional viewpoint as well. Recently, Hartline et al. [HJS24] studied a special case of online Bayesian ski rental problem in which $p_t = p$ for all days t . [HJS24] characterized an optimal worst-case online algorithm in the finite-horizon ski rental problem. The time horizon plays a crucial role in the decisions taken. For a large time horizon, specifically $T \gtrsim 2M$, the optimal strategy for the infinite-horizon ski rental problem [KMMO94] is optimal. The optimal worst-case algorithm is obtained by solving for equilibrium in a game that ignores the bad weather days. Solving for equilibrium in this restricted game is tractable. Moreover, [HJS24] proved that their online algorithm is also optimal in a special case of online Bayesian ski rental problem in which $p_t = p$ for all days t via a reduction to the worst-case algorithm problem. Diakonikolas et al. [DKT+21] consider a similar setting, in which they assume the good weather days are distributed according to a log-concave distribution. Crucially, however, the algorithm they propose depends on samples from the distributions. The problem we consider can be thought of as a zero-sample setting of [DKT+21].

Although the optimum online is a quite natural benchmark, it remains understudied. This is largely due to the technical challenge it presents compared to the optimum offline (see discussion in [Rou21, Section 24.4.1]).

1.3 Results and Further Work

As mentioned earlier, this note covers fundamental results regarding the polynomial-time approximability of the optimal online Bayesian bipartite matching algorithm (Section 2) and presents a new philosopher inequality for online Bayesian ski rental (Section 3). Section 2.1 contains description of the hardness results proved in [PPSW21] both for the weighted case and for the unweighted case of online Bayesian bipartite matching. Next, in Section 2.2, we describe the first algorithmic result (with approximation factor of 0.51) given in [PPSW21] and briefly explain how the current improved result (with approximation factor of 0.678) was obtained in [BDP+24]. Finally, in Section 3, we show a new linear program that captures the optimal online Bayesian ski rental algorithm *precisely*, in a similar fashion to the known result regarding the polynomial-time computability of the optimal online Bayesian bipartite matching with a single bin [PPSW21]. So it is easy to compute the optimal online algorithm for online Bayesian ski rental. However, only a 2-competitive prophet inequality is known for online Bayesian ski rental. This follows from a well-known result in computer science folklore, a 2-competitive deterministic algorithm for the adversarial variant of the ski rental problem.

Two directions for further work can be considered. First, a natural direction is to further study the efficient approximability of our two problems. More specifically, one can try to obtain better guarantees for both the approximation of the online Bayesian bipartite matching optimum and the approximation of the offline Bayesian ski rental optimum. Generally speaking, another essential direction is to explore how well one can approximate the optimal online algorithm of online Bayesian selection problems under the many constraints studied in the literature, including knap-

sack constraints, matroid constraints, etc. More precisely, for which of these problems is the online algorithm easy to compute? Which admit a PTAS? which admit constant approximations? Which are hard to approximate? The study of approximation of the optimal online algorithm is significant beyond than allowing for better quantitative bounds than achievable versus the offline optimal, as it opens avenues for more algorithmic and analytic for online resource allocation problems. Mechanism design considerations furthermore raise the question of what guarantees for the problem can be had while maintaining incentive compatibility.

2 Online Bayesian Bipartite Matching

2.1 Hardness Results

In this section we briefly depict how the following hardness result of approximation of the philosopher for the weighted version of the problem was obtained in [PPSW21].

Theorem 2.1 ([PPSW21]). *It is $PSPACE$ -hard to approximate the optimal online stochastic bipartite matching algorithm within a factor of α , for some absolute constant $\alpha < 1$. This remains true even when all weights and inverse arrival probabilities are bounded by some polynomial in the size of the input.*

To get this $PSPACE$ -hardness result, Papadimitriou et al. [PPSW21] first refined the result of Condon et al. [CFLS97] for maximum satisfiability of stochastic SAT instances. The stochastic SAT (SSAT) problem is defined as follows.

Definition 2.1 ([Pap85]). *An instance of the STOCHASTIC-3-SAT problem consists of a 3-CNF ϕ with variables $x_1, x_2, \dots, x_n$. At time $t = 1, 2, \dots, n$, if t is odd our (online) algorithm $\mathcal{A}$ chooses whether to set x_t to **True** or **False** whereas if t is even, nature sets x_t uniformly at random from $\{\text{True}, \text{False}\}$. The performance of our algorithm, $\mathcal{A}(\phi)$, is the expected number of satisfied clauses.*

Here, the maximization variant of the stochastic SAT (SSAT) problem is considered (named MAX-SSAT), in which the goal is to maximize the expected number of satisfied clauses in ϕ after all variables have been assigned a value. $\{x_1, x_3, \dots\}$ are referred as "deterministic variables" and $\{x_2, x_4, \dots\}$ are referred as "random variables". Condon et al. [CFLS97] proved the following.

Lemma 2.1 ([CFLS97]). *There exist constants $k \in \mathbb{N}$ and $\alpha \in (0, 1)$ so that it is $PSPACE$ -hard to compute an α -approximation to $OPT_{on}(\phi)$ for a MAX-SSAT instance ϕ satisfying:*

1. *no random variable appears negated in any clause of ϕ , and*
2. *each random variable appears in at most k clauses of ϕ .*

Using an *expander graph* construction, Papadimitriou et al. [PPSW21] extended the result of Lemma 2.1 to MAX-SSAT instances in which each variable appears in at most $O(1)$ clauses. More precisely, we define the following.

Definition 2.2. *The expansion of a graph G is defined as*

$$h(G) := \min_{S \subseteq V, |S| \leq |V|/2} \frac{|E(S, V \setminus S)|}{S},$$

where $E(X, Y) := \{(x, y) \in E \mid x \in X, y \in Y\}$ denotes the edges with one endpoint in X and the other in Y .

Papadimitriou et al. [PPSW21] utilized results providing explicit, deterministic constructions of graphs with constant degree and constant expansion (e.g. [GG81; LPS88]).

Lemma 2.2. *There exists a deterministic, polynomial-time construction of a graph on n vertices with expansion at least 1 and maximum degree at most some constant d .*

Using the construction of Lemma 2.2, the following extension of the result of Lemma 2.1 was shown in [PPSW21].

Lemma 2.3. *There exist constants $k \in \mathbb{N}$ and $\alpha \in (0, 1)$ so that it is **PSPACE**-hard to compute an α -approximation to $OPT_{on}(\phi)$ for a MAX-SSAT instance ϕ satisfying:*

1. *no random variable appears negated in any clause of ϕ , and*
2. *each variable (both random and deterministic) appears in at most k clauses of ϕ .*

Then, Papadimitriou et al. [PPSW21] gave a polynomial-time reduction from approximating maximum satisfiability of a bounded-occurrence SSAT instance to approximating the optimal online algorithm for the online stochastic bipartite matching problem, yielding the claimed **PSPACE**-hardness result (Theorem 2.1).

2.1.1 The Unweighted Case

We now consider the unweighted version of the problem and describe the following hardness result of approximation of the philosopher proved in [BDP+24].

Theorem 2.2 ([BDP+24]). *It is **PSPACE**-hard to approximate OPT_{on} within some universal constant $\alpha < 1$, even for unweighted Bernoulli instances with arrival probabilities bounded away from zero.*

To prove this **PSPACE**-hardness result, the hardness result of approximation for the bounded-degree stochastic SAT problem showed in [PPSW21] was considered (Lemma 2.3). The following novel contribution implies that such a reduction works even without edge weights or small arrival probabilities.

Consider a STOCHASTIC-3-SAT instance ϕ with variables $\{x_1, x_2, \dots, x_n\}$ that satisfies the conditions in Lemma 2.3. We construct a corresponding online stochastic bipartite matching instance $\mathcal{I}_\phi$ which first has n arrivals called "variable nodes" corresponding to $\{x_1, x_2, \dots, x_n\}$. For odd $t \in [n]$, the corresponding variable node arrives with probability 1 and neighbors two bins $\{T^t, F^t\}$. For even $t \in [n]$, the corresponding variable node arrives with probability $1/2$, and neighbors exactly one bin $\{F^t\}$.

The instance $\mathcal{I}_\phi$ then consists of stochastic "clause nodes" all arriving with a common probability $p \leq 0.1$. For each clause C of ϕ , there is a corresponding stochastic arrival of degree at most 3. The neighborhood of the stochastic arrival corresponding to C is constructed as follows: for each of the literals $l_i \in C$, if $l_i = x_i$ we add an edge to F^i and if $l_i = \neg x_i$ we add an edge to T^i ¹.

There is a natural bijection between algorithms for the STOCHASTIC-3-SAT instance ϕ , and stochastic bipartite matching algorithms for $\mathcal{I}_\phi$ which match every arriving variable node. As optimum online algorithm is one such algorithm, in [BDP+24, Appendix F] it was shown that its performance on $\mathcal{I}_\phi$ is bounded in terms of $OPT_{on}(\phi)$. However, because bins have low degrees, it was shown that the amount of noise of this introduces is of order p^2 , while the signal that relates $OPT_{on}(\mathcal{I}_\phi)$ to $OPT_{on}(\phi)$ is of order p . Consequently, taking p to be sufficiently small constant suffices to obtain the desired hardness result of approximation (Theorem 2.2).

¹Note that it is always possible by the assumption that ϕ has no randomly set variable appear negated in any clause.

2.2 Algorithmic Results

In this section we describe the first algorithm for approximating the profit of OPT_{on} (with approximation factor of 0.51) given in [PPSW21].

Theorem 2.3 ([PPSW21]). *There exists a polynomial-time online algorithm which is a 0.51-approximation of the optimal online algorithm for the online Bayesian bipartite matching problem.*

The core of the algorithm is a linear program (LP) called LP-match, for which it was shown that it upper bounds the profit of any online algorithm for online Bayesian bipartite matching. In particular, denoting by $LP\text{-}Match(\mathcal{I})$ the optimal value of LP-Match on Instance $\mathcal{I}$, the following lemma was shown.

Lemma 2.4. *For any online Bayesian bipartite matching instance $\mathcal{I}$, we have that*

$$LP\text{-}Match(\mathcal{I}) \geq OPT_{on}(\mathcal{I}).$$

In addition, LP-Match precisely captures the optimal online algorithm for instances with a single bin i (see [PPSW21, Appendix C]). Below, the optimizing variables are $\{x_{i,t}\}$, which indicate "the probability that the online algorithm matches ball t to bin i ". Recall that ball t arrives with probability p_t .

$$\text{LP-Match:} \quad \text{maximize} \quad \sum_{i,t} w_{i,t} \cdot x_{i,t} \quad (1a)$$

$$\text{subject to} \quad \sum_t x_{i,t} \leq 1 \quad \text{for all } i, \quad (1b)$$

$$\sum_i x_{i,t} \leq p_t \quad \text{for all } t, \quad (1c)$$

$$x_{i,t} \leq p_t \cdot \left(1 - \sum_{t' < t} x_{i,t'}\right) \quad \text{for all } i, t, \quad (1d)$$

$$x_{i,t} \geq 0 \quad \text{for all } i, t. \quad (1e)$$

The Algorithm. Given a solution to LP-Match, a natural approach to approximate OPT_{on} is to round this solution online. By simple "integrality gap" examples, this is impossible to do perfectly (see [PPSW21, Appendix C]). Instead, this was done approximately, by rounding a solution to LP-Match while only incurring a $1/2 + c$ multiplicative loss in the rounding, for the constant $c := 0.01$. Also, for notational simplicity, we assume without loss of generality that an optimal solution to LP-Match to the input instance $\mathcal{I}$ satisfies all Constraints (1c) at equality. This can be guaranteed by adding a dummy bin i_t for each ball t with $w_{i_t,t} = 0$, and setting $x_{i_t,t} \leftarrow p_t - \sum_i x_{i,t}$. These dummy edges do not affect the profit of OPT_{on} , nor that of the online algorithm. Below is the algorithm given in [PPSW21].

Algorithm 1 Rounding LP-Match Online

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1: solve LP-Match for  $\{x_{i,t}\}$ 
2: add dummy neighbor for each  $t$  so that  $\sum_i x_{i,t} = p_t$ 
3:  $\mathcal{M} \leftarrow \emptyset$ 
4: for all balls  $t = 1, 2, \dots$  do
5:   pick a single bin  $i$  with probability  $\frac{x_{i,t}}{p_t}$ 
6:   if  $i$  is unmatched in  $\mathcal{M}$  then
7:     with probability  $q_{i,t} := \min\left(1, \frac{1/2+c}{1-\sum_{t'<t} x_{i,t'} \cdot (1/2+c)}\right)$  do  $\mathcal{M} \leftarrow \mathcal{M} \cup \{(i, t)\}$ 
8:   if  $t$  is still unmatched in  $\mathcal{M}$  then
9:     pick a single bin  $i$  with probability  $\frac{x_{i,t}}{p_t}$ 
10:    if  $i$  is unmatched in  $\mathcal{M}$  and  $\sum_{t'<t} x_{i,t'} > \frac{1/2-c}{1/2+c}$  then
11:       $\mathcal{M} \leftarrow \mathcal{M} \cup \{(i, t)\}$ 
12: Output  $\mathcal{M}$ 
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Clearly, the algorithm outputs a matching. Also, the following per-edge guarantees for Algorithm 1 are shown.

Theorem 2.4. *Each edge $(i, t) \in E$ is matched by Algorithm 1 with probability at least*

$$\Pr[(i, t) \in \mathcal{M}] \geq x_{i,t} \cdot (1/2 + c).$$

Note that Theorem 2.3 implies that Algorithm 1 is a polynomial-time 0.51-approximation of the optimal online algorithm, thus proving Theorem 2.3.

Proof of Theorem 2.3. All steps of Algorithm 1, including solving the polynomially-sized LP in Line 1, can be performed in polynomial time. The approximation ratio follows directly from linearity of expectation together with Lemma 2.4 and Theorem 2.4. ■

The rest of the analysis was dedicated to proving Theorem 2.4. To do so, two events for edge (i, t) being match were considered- depending on whether it was matched as a first pick (Line 7) or second pick (Line 11). In both the analysis for first pick (Line 7) and the analysis for second pick (Line 11) the following definition is used.

Definition 2.3. *Ball t is early for bin i if $\sum_{t'<t} x_{i,t'} \leq \frac{1/2-c}{1/2+c}$. Otherwise, it is late. Edge (u, t) is early (late) if t is early (late) for bin i .*

Note that by definition early edges cannot be matched as a second pick (Line 11). So in the analysis for second pick (Line 11) only late edges have to be considered. The next lemma was proved for bounding the probability of an edge (i, t) being matched as a first pick (Line 7).

Lemma 2.5. *If edge $(i, t) \in E$ is early, then*

$$\Pr[(i, t) \in \mathcal{M} \text{ as a first pick}] = x_{i,t} (1/2 + c).$$

In addition, for any edge $(i, t) \in E$,

$$x_{i,t} \cdot (1/2 - 3c) \leq \Pr[(i, t) \in \mathcal{M} \text{ as a first pick}] \leq x_{i,t} \cdot (1/2 + c).$$

In the analysis for a second pick (Line 11), the next lemma was proved which is sufficient to prove Theorem 2.4 given Lemma 2.5.

Lemma 2.6. *For any late edge $(i, t) \in E$,*

$$\Pr[(i, t) \in \mathcal{M} \text{ as a second pick}] \geq x_{i,t} \cdot 4c.$$

Generalizing the Algorithm. Algorithm 1 and its analysis generalize seamlessly to a setting in which weights of each ball t are drawn from discrete joint distributions. In fact, if each of the discrete joint distributions consists of two-point distributions, with one of the possible realizations of the weights of the ball being all zeros, we capture exactly the problem we discussed. See [PPSW21, Section 5] for the outline of the necessary changes in the LP, algorithm and analysis to generalize the algorithm.

2.2.1 The Improvement

In this section we briefly explain how the current improved result of approximating the profit of OPT_{on} (with approximation factor of 0.678) was obtained in [BDP+24]. First, Braverman et al. [BDP+24] observed that LP-Match without Constraints (1b) still precisely captures the optimal on-line algorithm for instances with a single bin i . More specifically, Constraints (1d)-(1e) are sufficient to match each edge with probability precisely $x_{i,t}$ by matching with probability $\frac{x_{i,t}}{p_t(1 - \sum_{t' < t} x_{i,t'})}$ if i is still unmatched and t arrives. Performing this in some sense independently for each bin i can be viewed similarly to the approach of the $(1 - 1/e)$ -approximate algorithm of [BDM22], to obtain "proposal" from bins, and then matching t to the highest-valued neighbor. For this approach a simple bound of $1 - 1/e$ was obtained. Essentially, this bound is tight, as it is possible to construct tight examples, even in unweighted instances (e.g., for a single ball t with $p_t = 1$ and m neighbors and LP-Match (without Constraints (1b)) solution $x_{i,t} = \frac{1}{m}$). The improvement on this algorithm is obtained by maintaining the same marginals while *correlating* the proposals and careful scaling, thus guaranteeing fewer collisions. The current improvement of [BDP+24] was achieved by correlating these proposals *optimally*, via a classical offline algorithm: (linear-order) *pivotal sampling* [Sri01; GKPS06]. See [BDP+24] for the improved algorithm itself and its analysis. Finally, it is important to note that the improved algorithm and its analysis can also be extended to the more general setting mentioned earlier (as noted in [BDP+24, Appendix G]).

3 Online Bayesian Ski Rental

In this section we give an algorithm to compute the cost of OPT_{on} for the online Bayesian ski rental problem.

Theorem 3.1. *There exists a polynomial-time online algorithm which captures the optimal online Bayesian ski rental algorithm precisely.*

An LP Relaxation. We first formulate a linear program (LP) called LP-Rent, which we show lower bounds the cost of any online Bayesian ski rental algorithm. Below, the variables we optimize over are $\{x_t\}$ and $\{z_t\}$, which we think as "the probability that the online algorithm buys the skis

on day t " and "the probability that the online algorithm rents the skis on day t ", respectively.

$$\textbf{LP-Rent:} \quad \text{minimize} \quad M \sum_t x_t + \sum_t z_t \quad (2a)$$

$$\text{subject to} \quad \sum_t x_t \leq 1, \quad (2b)$$

$$x_t \leq p_t \cdot \left(1 - \sum_{i < t} x_i\right) \quad \text{for all } t, \quad (2c)$$

$$z_t \leq p_t \cdot \left(1 - \sum_{i < t} x_i\right) \quad \text{for all } t, \quad (2d)$$

$$x_t + z_t \geq p_t \cdot \left(1 - \sum_{i < t} x_i\right) \quad \text{for all } t, \quad (2e)$$

$$x_t \geq 0 \quad \text{for all } t, \quad (2f)$$

$$z_t \geq 0 \quad \text{for all } t. \quad (2g)$$

Given a online Bayesian ski rental instance, the algorithm of Theorem 3.1 simply computes an optimal solution for LP-Rent and acts as the online algorithm in the proof of Lemma 3.2. Clearly, this is a polynomial-time online algorithm because writing and solving LP-Rent for any Bayesian ski rental instance can be done in a polynomial time. Let us denote by $\text{LP-Rent}(\mathcal{I})$ the optimal value of LP-Rent on instance $\mathcal{I}$. We have the following lemma.

Lemma 3.1. *For any online Bayesian ski rental instance, we have that*

$$\text{LP-Rent}(\mathcal{I}) \leq \text{OPT}_{on}(\mathcal{I}).$$

Proof. Let x_t^* (z_t^*) denote the probability that OPT_{on} buys (rents) the skis on day t . Notice that x^* and z^* constitute a feasible solution for LP-Rent because (i) the probability that OPT_{on} buys the skis is at most 1, (ii) both the probability that OPT_{on} buys the skis on day t and the probability that OPT_{on} rents the skis on day t are at most p_t (the probability that t is a good weather day) times $1 - \sum_{i < t} x_i$ (the probability that the skis were not bought by day t)², (iii) the probability that OPT_{on} buys or rents the skis on day t (this is the probability since the events are disjoint) is at least p_t (the probability that t is a good weather day) times $1 - \sum_{i < t} x_i$ (the probability that the skis were not bought by day t), and (iv) these probabilities are non-negative. On the other hand, for these x^* and z^* , the objective of LP-Rent (2a) is precisely the expected cost of OPT_{on} on this instance, and therefore $\text{LP-Rent}(\mathcal{I}) \leq \text{OPT}_{on}(\mathcal{I})$. ■

To prove Theorem 3.1 it is sufficient by Lemma 3.1 to show the following.

Lemma 3.2. *For any online Bayesian ski rental instance, we have that*

$$\text{LP-Rent}(\mathcal{I}) \geq \text{OPT}_{on}(\mathcal{I}).$$

Proof. Consider the following online algorithm, which starts by computing a solution $\vec{x}, \vec{z}$ to LP-Rent. Next, if day t arrives (as a good weather day) and the skis are still available to buy or rent,

²This uses the fact that once the skis are bought there is no need to buy or rent them, and the fact that the arrival of t is independent of the online algorithm's previous choices. Note that these constraints are not valid for probabilities induced by an *offline* algorithm, so LP-Rent does not lower bound $\text{OPT}_{off}(\mathcal{I})$.

buys the skis on day t with probability

$$q_t^b := \frac{x_t}{p_t (1 - \sum_{i < t} x_i)},$$

or rents the skis on day t with probability

$$q_t^r := \frac{z_t}{p_t (1 - \sum_{i < t} x_i)}.$$

These two quantities are indeed probabilities, by Constraints (2c) and Constraints (2d). For every day t , let B_t be the indicator random variable for the event that the skis are bought on day t . Let R_t be the indicator random variable for the event that the skis are rented on day t . Finally, let V_t be the indicator random variable for the event that the skis are *vacant* (i.e., available to buy or rent) on day t . We can use the following claim (which we prove later by induction):

Claim 3.1. *For each t , we have that $\Pr[B_t] = x_t$, $\Pr[R_t] = z_t$, and consequently $\Pr[V_t] = 1 - \sum_{i < t} x_i$.*

By linearity of expectation, Claim 3.1 implies that the online algorithm for instance $\mathcal{I}$ has expected cost precisely

$$B \sum_t x_t + \sum_t z_t = \text{LP-Rent}(\mathcal{I}).$$

Consequently, $\text{LP-Rent}(\mathcal{I}) \geq \text{OPT}_{on}(\mathcal{I})$. ■

Proof of Claim 3.1. We prove by strong induction that these equations hold for all days t .

Base Case $t = 1$: Consider the equations where $t = 1$. Clearly, we have that

$$\Pr[V_1] = 1. \tag{3}$$

Now, the event that the skis are bought on day 1 requires that day 1 arrives (as a good weather day), that the skis are vacant on day 1, and that the skis are selected to buy on day 1. Note that the skis being vacant on day 1 is independent from the arrival of 1, and the selection to buy the skis on day 1. Therefore,

$$\Pr[B_1] = p_1 \cdot q_1^b \cdot \Pr[V_1] = p_1 \cdot \frac{x_1}{p_1} \cdot 1 = x_1,$$

where the second equality uses Equation (3). Similarly, the event that the skis are rent on day 1 requires that day 1 arrives (as a good weather day), that the skis are vacant on day 1, and that the skis are selected to rent on day 1. Note that the skis being vacant on day 1 is independent from the arrival of 1, and the selection to rent the skis on day 1. Therefore,

$$\Pr[R_1] = p_1 \cdot q_1^r \cdot \Pr[V_1] = p_1 \cdot \frac{z_1}{p_1} \cdot 1 = z_1.$$

where the second equality uses Equation (3).

Inductive Step: Assuming that the claim holds for all $i < t$, we will prove it holds for t as well. First, we have that

$$\Pr[V_t] = 1 - \sum_{i < t} \Pr[B_i] = 1 - \sum_{i < t} x_i, \tag{4}$$

where the last equality follows from the inductive hypothesis. Now, the event that the skis are bought on day t requires that day t arrives (as a good weather day), that the skis are vacant on day t , and that the skis are selected to buy on day t . Note that the skis being vacant on day t is independent from the arrival of t , and the selection to buy the skis on day t . Therefore,

$$\Pr[B_t] = p_t \cdot q_t^b \cdot \Pr[V_t] = p_t \cdot \frac{x_t}{p_t (1 - \sum_{i < t} x_i)} \cdot \left(1 - \sum_{i < t} x_i\right) = x_t,$$

where the second equality uses Equation (4). Similarly, the event that the skis are rent on day t requires that day t arrives (as a good weather day), that the skis are vacant on day t , and that the skis are selected to rent on day t . Note that the skis being vacant on day t is independent from the arrival of t , and the selection to rent the skis on day t . Therefore,

$$\Pr[R_t] = p_t \cdot q_t^r \cdot \Pr[V_t] = p_t \cdot \frac{z_t}{p_t (1 - \sum_{i < t} x_i)} \cdot \left(1 - \sum_{i < t} x_i\right) = z_t,$$

where the second equality uses Equation (4). Hence the proof. ■

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